

Freight Rate Prediction

ML Engineering Assessment — SpotterLabs

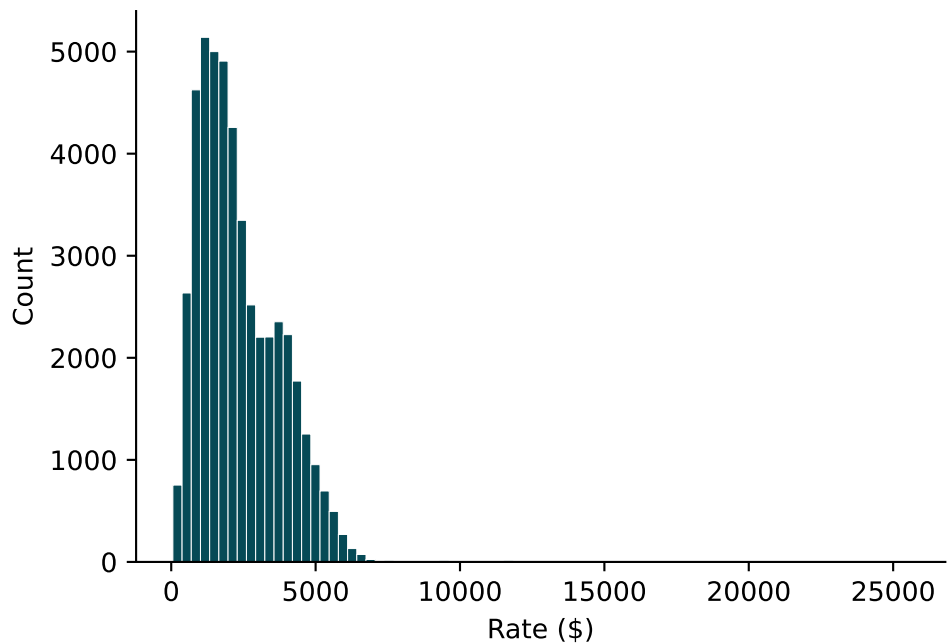
Muhammad Moazzam

Model: LightGBM + XGBoost Ensemble | Hold-out RMSE: \$639 | MAPE: 6.1%

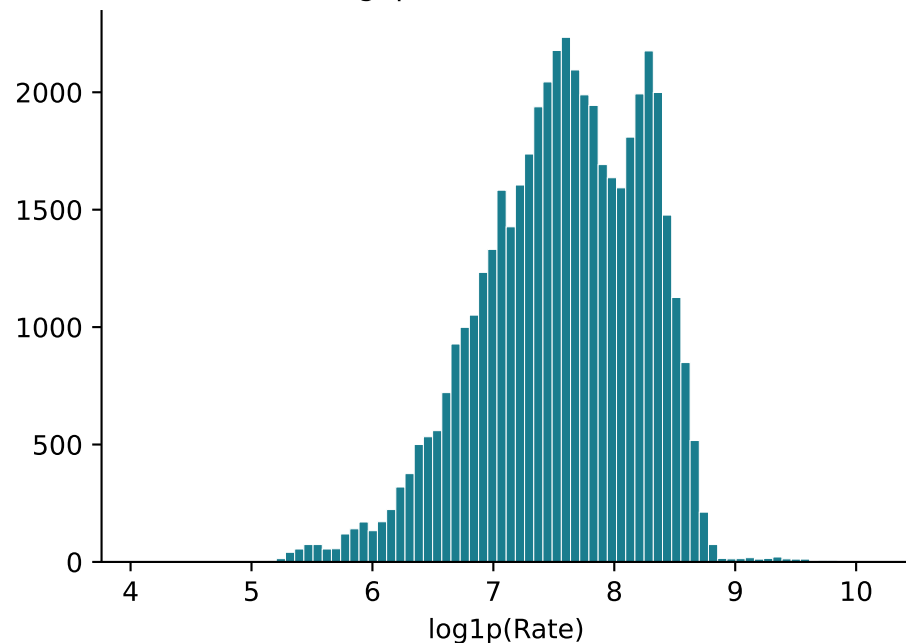
1. Data Overview

Train: 48,000 rows | Jan-Oct 2025 | 14 features | Rates 57 – 25,533 | Missing: weight(0.6%), market_index(0.8%)

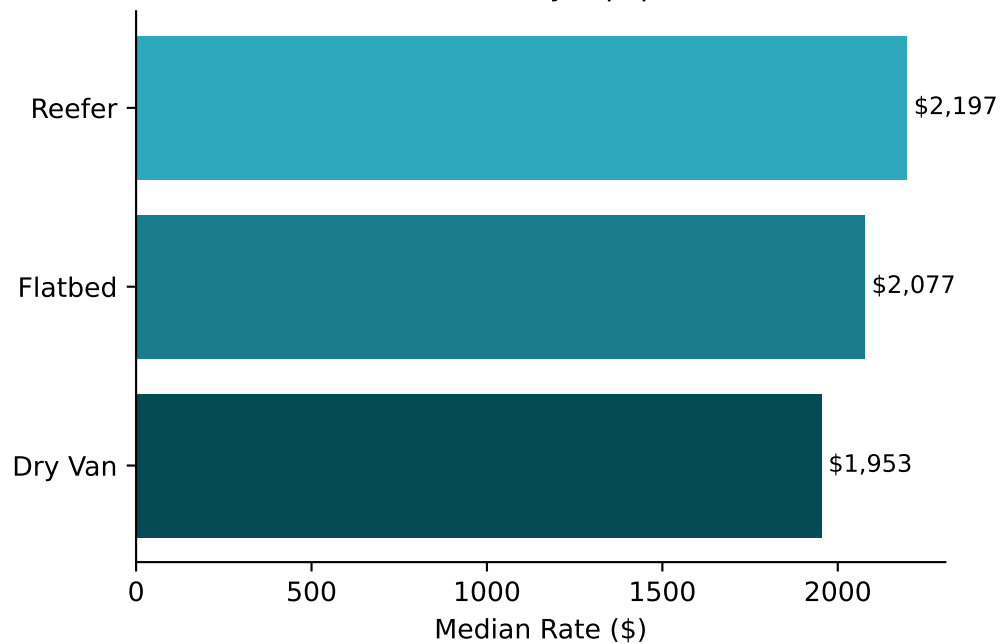
Rate distribution (raw)



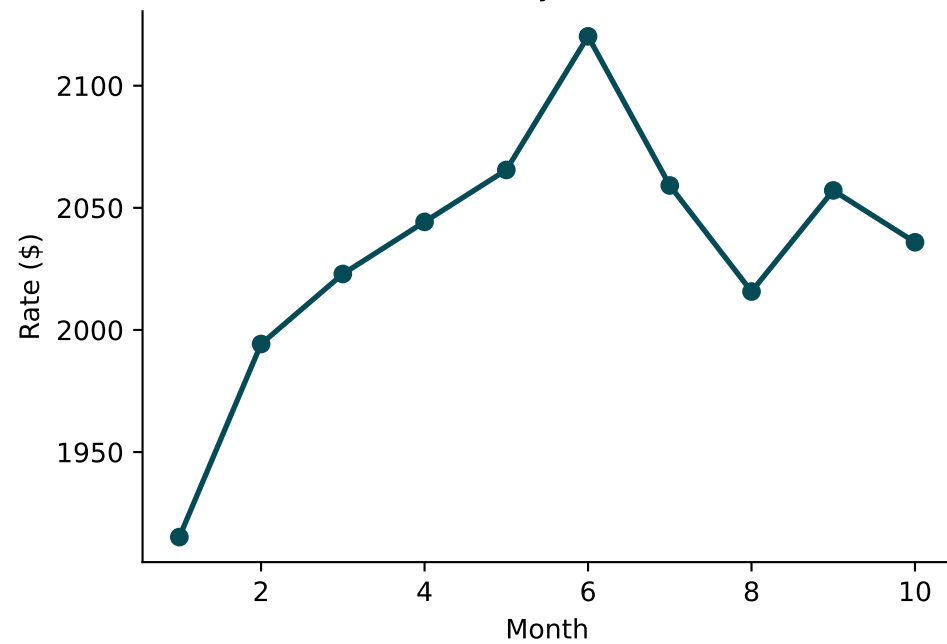
log1p(Rate) distribution



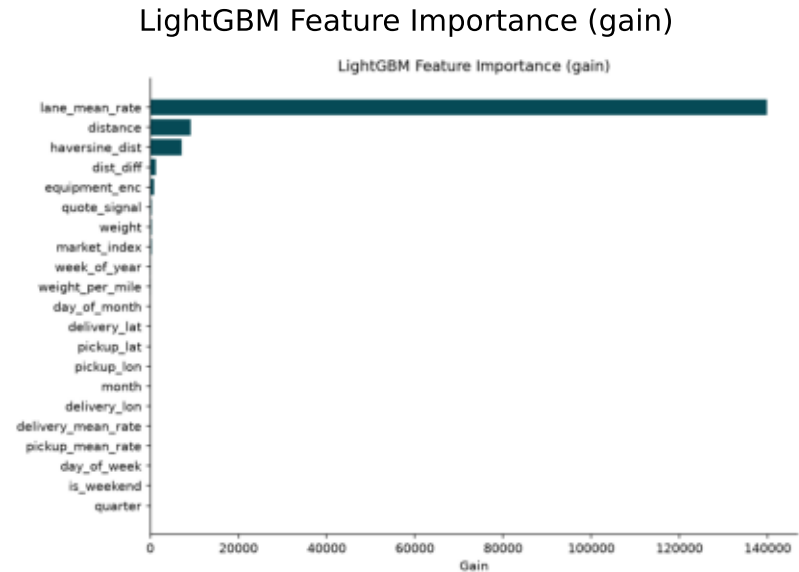
Median rate by equipment



Median rate by month (train)



2. Data Split Strategy & Feature Engineering



Time-Based Split Rationale

- Training set : Jan–Sep 2025 (43,147 rows)
- Hold-out set : Oct 2025 (4,853 rows)
- Final test : Nov–Dec 2025 (12,000 rows)

Using a temporal split (not random) prevents data leakage and simulates real deployment: the model only ever sees past data when predicting future loads.

Features Engineered

Date features

day_of_week, month, week_of_year,
day_of_month, is_weekend, quarter

Load features

weight_per_mile = weight / distance
haversine_dist (from lat/lon)
dist_diff = stated - haversine

City encoding

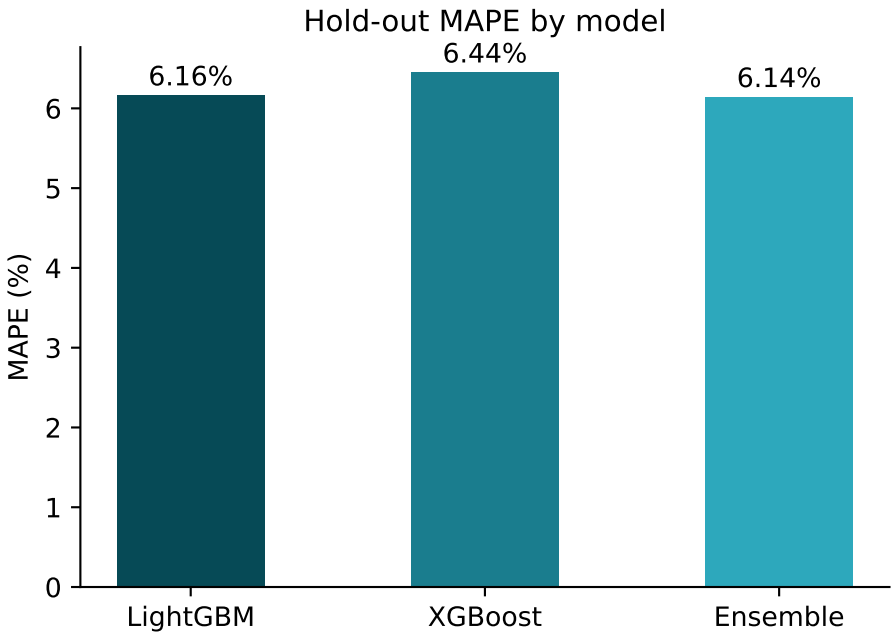
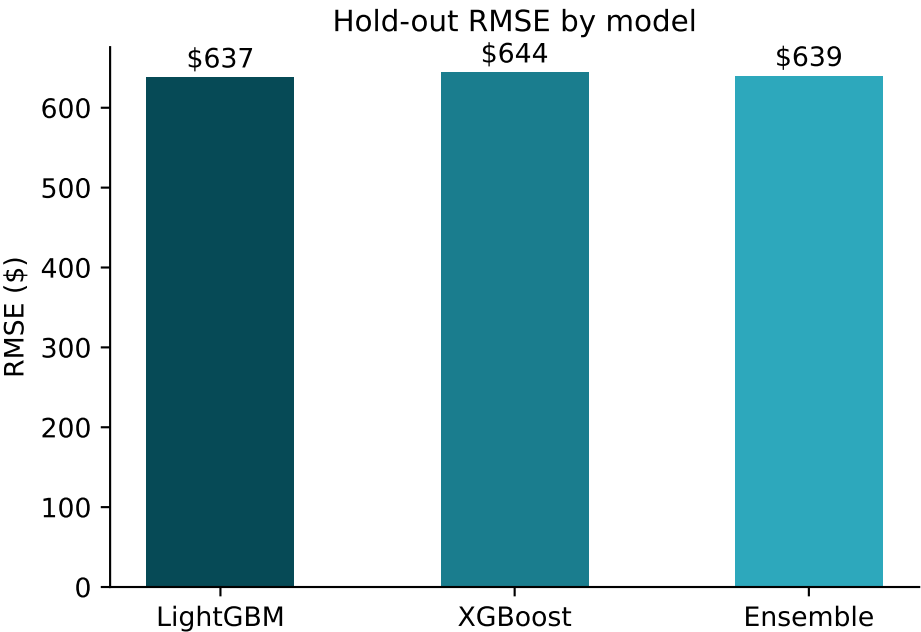
Smoothed target-mean for pickup / delivery
Lane-level mean rate (pickup+delivery pair)

Pass-through

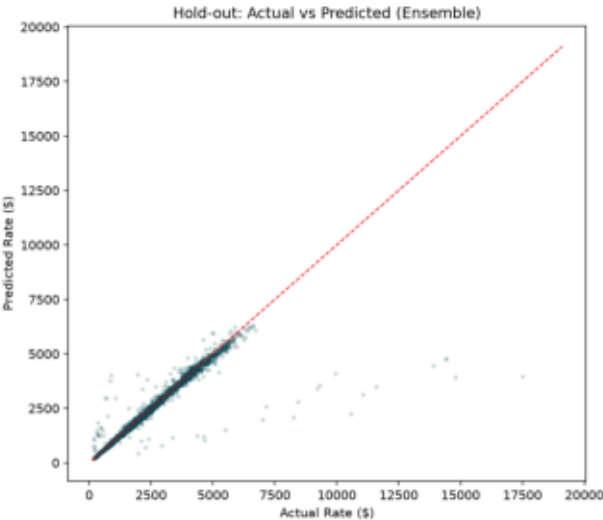
market_index, quote_signal <- top signals
distance, weight, equipment_enc

3. Model Training & Validation Results

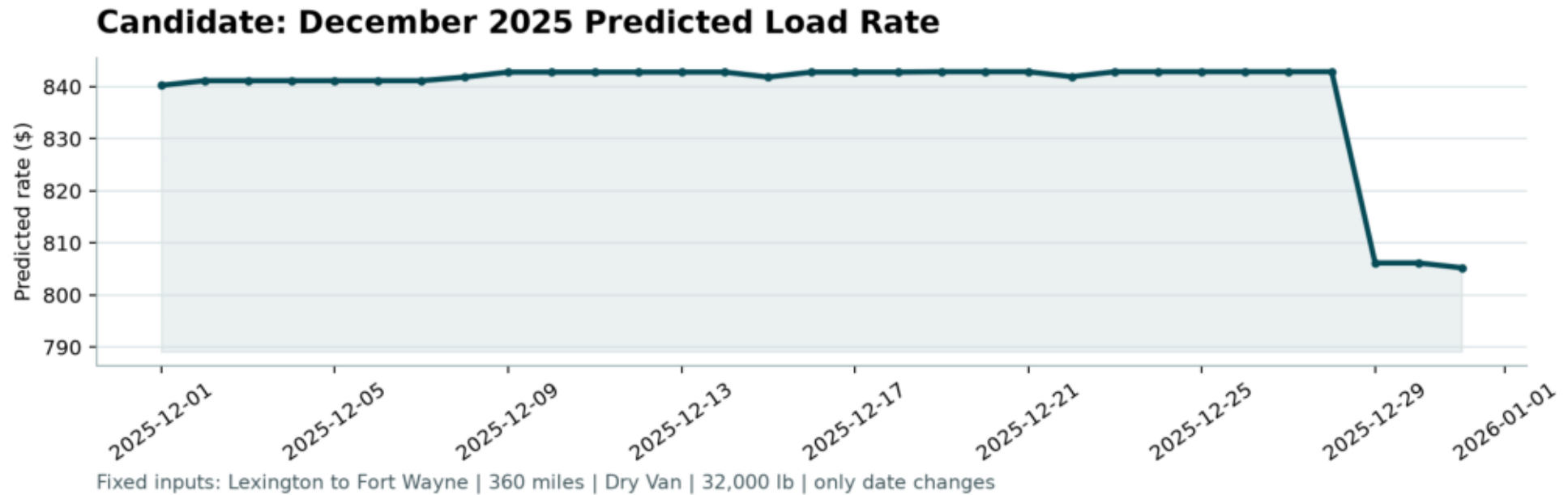
Primary: LightGBM (93 trees, lr=0.05, leaves=127) | Secondary: XGBoost (78 trees, lr=0.05, depth=7) | Ensemble: inverse-RMSE weighted average | Target: log1p(rate)



Hold-out: Actual vs Predicted (Ensemble)



4. December 2025 Predicted Load Rate Chart



Fixed lane: Lexington -> Fort Wayne | 360 mi | Dry Van | 32,000 lb | market_index and quote_signal set to training medians for Dec 2025 inference.

Observation: Rates are stable ~\$840 through Dec 25, then drop to ~\$805 Dec 28-31 — consistent with reduced freight demand during the holiday week.